

Smart Wireless Network Traffic Guardian

Problem Statement

5G networks experience congestion due to uneven traffic distribution, rapid capacity variation, and reactive load balancing mechanisms. Existing control methods cannot adapt fast enough to dynamic user demand and mobility. This results in overloaded cells, underutilized resources, increased latency, and reduced Quality of Service. There is a need for an intelligent load balancing control system that can predict traffic patterns and dynamically redistribute network load in real time to minimize congestion and optimize resource utilization.

Objectives

Main Objective

Develop a fully functional AI-based wireless network congestion prediction and traffic optimization system that can be safely demonstrated and validated in a controlled environment, with potential for future deployment in large networks.

Specific Objectives

1. **Define system scope and requirements** – Identify key congestion metrics, target network scale, and safe test conditions for a controlled wireless environment.
2. **Set up a controlled test network** – Configure a personal router and connected devices to simulate realistic wireless traffic, including peak loads and user mobility patterns.
3. **Collect and preprocess data** – Record network performance metrics such as device count, throughput, latency, packet loss, and channel utilization under normal and congested conditions.
4. **Develop congestion prediction model** – Build and train a machine learning model that forecasts network congestion before performance degrades.
5. **Design load optimization module** – Create algorithms that suggest traffic management actions, such as client steering, channel reassignment, or bandwidth allocation.

6. **Integrate prediction and optimization** – Combine the AI model and traffic management module into a single decision-support system.
7. **Build monitoring and visualization dashboard** – Provide a clear interface showing predicted congestion and recommended actions for demonstration purposes.
8. **Test and validate system** – Evaluate prediction accuracy, load distribution efficiency, latency, throughput, and packet loss under controlled conditions.
9. **Demonstrate safe prototype to university** – Show the system's predictions and recommendations without connecting to the live campus network.
10. **Document design and future deployment potential** – Prepare full system documentation including methodology, assumptions, limitations, and steps to scale to larger networks or real 5G deployment.

Methodology

Phase 1: Requirements Analysis and System Planning

1. **Define project scope** – Identify key goals: congestion prediction, traffic optimization, safe demonstration.
2. **Identify metrics** – Select measurable KPIs: device count per access point, throughput, latency, packet loss, channel utilization.
3. **Determine test network setup** – Choose router hardware, number of connected devices (simulate users), and traffic patterns for controlled experiments.
4. **Plan data collection strategy** – Decide which data to collect, sampling frequency, and storage format for analysis and model training.

Phase 2: Controlled Network Setup

1. **Install and configure router** – Use a router that supports monitor
2. **Connect devices** – Simulate multiple users with smartphones, laptops, or IoT devices.
3. **Introduce traffic loads** – Create realistic traffic using streaming, downloads, or scripted network activity. Include high-load scenarios to simulate congestion.

4. **Verify network monitoring** – Ensure metrics such as throughput, latency, packet loss, and device counts are correctly logged.

Phase 3: Data Collection and Preprocessing

1. **Record network metrics** – Collect data during normal and congested periods.
2. **Clean and preprocess data** – Remove anomalies, normalize features, handle missing data.
3. **Feature engineering** – Calculate additional parameters like utilization ratio, average throughput per device, and latency growth trends.
4. **Label congestion events** – Define thresholds to mark periods of congestion

Phase 4: AI Model Development

1. **Select model type** – Start with lightweight models suitable for small-scale deployment: regression, decision trees, or simple neural networks.
2. **Train the model** – Use historical and real-time collected data to train the model to predict congestion within 5–15 minutes ahead.
3. **Validate the model** – Test prediction accuracy using a holdout dataset or cross-validation.
4. **Refine model** – Adjust parameters, features, or algorithm type to improve accuracy and reduce false predictions.

Phase 5: Load Optimization Algorithm Design

1. **Define actions** – Identify possible network interventions: client steering, channel switching, traffic shaping, or bandwidth prioritization.
2. **Algorithm development** – Create rules or optimization logic that recommends actions based on predicted congestion.
3. **Simulation of interventions** – Test the algorithm in the controlled network to see predicted performance improvements.

Phase 6: System Integration

1. **Combine AI model and optimization module** – Ensure predictions feed directly into decision recommendations.
2. **Build decision-support interface** – Develop a dashboard showing predicted congestion, recommended actions, and KPI trends.
3. **Implement logging and monitoring** – Capture real-time system behavior and predicted vs actual congestion events.

Phase 7: Testing and Validation

1. **Simulate multiple traffic scenarios** – Include normal load, peak load, and extreme congestion to evaluate performance.
2. **Measure KPIs** – Latency, throughput, packet loss, and prediction accuracy.
3. **Compare with baseline** – Evaluate system performance against a simple rule-based or static threshold method.
4. **Iterate** – Refine AI model and optimization algorithm based on test results.

Phase 8: Safe Demonstration to University

1. **Prepare demo plan** – Show predictions and recommended actions only; do not interfere with live campus network.
2. **Use controlled network** – Demonstrate system on your personal router setup with simulated users.
3. **Highlight performance improvements** – Use graphs and metrics to show congestion prediction accuracy and effectiveness of suggested interventions.

Phase 9: Documentation and Reporting

1. **Document system design** – Include architecture, AI model details, optimization logic, and dashboard screenshots.
2. **Detail methodology** – Explain data collection, feature selection, model training, and testing procedures.

3. **Analyze limitations** – Discuss scaling to larger networks and potential deployment in real 5G networks.
4. **Provide future recommendations** – Steps to integrate with private 5G networks or enterprise WiFi deployments.

Phase 10: Final Review and Deliverables

1. Deliver a fully functional prototype on your test network.
2. Include the dashboard for monitoring predictions and recommended traffic management actions.
3. Submit complete documentation, methodology, and performance evaluation results.

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Timeline

Timeline and Cost Plan: AI Wireless Congestion System

Phase	Period	Tasks	Estimated Cost (R)
1. Requirements Analysis & Planning	Feb 12 – Feb 28	Define scope, KPIs, test network requirements, data collection plan	0 (self-study, meetings)
2. Controlled Network Setup	Mar 1 – Mar 31	Buy router, configure network, connect devices, verify monitoring	4,000 – 6,000 (router + extra devices if needed)
3. Data Collection & Preprocessing	Apr 1 – Apr 30	Simulate traffic, collect metrics, clean and label data	0 – 1,000 (software, storage, scripts)
4. AI Model Development	May 1 – May 31	Train ML model, validate, refine, feature engineering	0 – 1,500 (MATLAB/Python packages, compute)

Phase	Period	Tasks	Estimated Cost (R)
5. Load Optimization Algorithm	Jun 1 – Jun 15	Design traffic redistribution logic, test in simulation	0
6. System Integration	Jun 16 – Jul 15	Combine prediction and optimization modules, build dashboard	500 – 1,000 (dashboard software, visualization tools)
7. Testing & Validation	Jul 16 – Aug 15	Simulate scenarios, measure KPIs, compare with baseline, refine	0 – 500 (additional scripts, test devices if needed)
8. Safe University Demo	Aug 16 – Sep 15	Prepare demo, run controlled demonstration, collect feedback	0
9. Documentation & Reporting	Sep 16 – Sep 25	Complete methodology, design, evaluation, and future recommendations	0
10. Final Review & Submission	Sep 26 – Sep 30	Compile final deliverables and prepare presentation	0

Total Estimated Cost: ~R5,000 –R 9,000